

Weekly Progress Report

FADO: Fairness-Aware Online Learning Under Concept Drift

Week ending 3 July 2026

Summary

Good news this week: I extended FADO to **multiple protected attributes** (intersectional fairness) — both the code and the experiments to back it up. The FOLKTABLES intersectional sweep is done, and the payoff is nice: the drift-reaction controller now delivers a *statistically significant* fairness gain over the controller-free baseline — something the old single-attribute setting was never able to show. The COMPAS sweep isn't finished yet, for a slightly unusual reason (see the note below).

A small logistical hiccup — experiments on pause

Confession: my machines and the current heatwave are not on speaking terms. With no air conditioning at home, running the PCs at full load for hours turns my room into a small sauna, and the hardware isn't thrilled either. So sweeps and experiments are **on hold** until either weather cools down (maybe). Everything already finished (the FOLKTABLES results below) is safe and sound — it's only the pending runs that are waiting for a less tropical week.

1. Multiple protected attributes (intersectional fairness)

Motivation and design decision

Previously FADO enforced fairness on a single binary attribute. We extended it to enforce fairness jointly over **two** attributes. There were two candidate designs:

- **Path A – Intersectional.** Optimise demographic parity (DP) over the Cartesian-product subgroups: the joint group index is $a = 2a^{(1)} + a^{(2)} \in \{0, 1, 2, 3\}$ ($K = 4$ groups).
- **Path B – Marginal sum.** Regularise each attribute independently.

We adopted **Path A**. Kearns et al. (ICML 2018, *Preventing Fairness Gerrymandering*) prove that marginal-only regularisation is *provably defeatable*: a model can achieve perfect per-attribute parity while being arbitrarily unfair on the joint subgroups. Path A closes this loophole by construction, aligns with the dominant intersectional-fairness definition (Foulds et al., ICDE 2020), and — importantly — is also the smaller code change, since the node-level regulariser auto-scales to any K .

Implementation

- **Data loaders:** composite group index built in the loaders — COMPAS uses `race × sex`, FOLKTABLES uses `SEX × RAC1P` (binarised). Row-alignment invariants preserved through the per-seed subsampling and the COMPAS drift edits.
- **Metrics:** the intersectional DP (over the four joint cells) is the optimised objective; the two *marginal* per-attribute DP values are computed and stored as diagnostics for every run.
- **Learning algorithm unchanged:** the fair oblique forest, the dual-ADWIN detector, and the reaction controller required *no* modification — they operate on whatever number of groups the data provides.

2. Experiments

All runs use the prequential (test-then-train, batch size 1) protocol, 30 random seeds, paired Wilcoxon signed-rank tests with Holm–Bonferroni correction.

- **Folktables (ACS Income, CA), sex $\times$ race — COMPLETE.** 30 seeds $\times$ 4 models (FADO, controller-free Aranyani, ARF, RFR).
- **COMPAS, race $\times$ sex — PAUSED** (see availability note); the pipeline is implemented and validated on single seeds, only the full 30-seed sweep is outstanding.

3. Results (Folktables, intersectional DP)

Mean $\pm$ std over 30 seeds. DP is the intersectional violation over the four joint subgroups (the objective FADO optimises); DP_{sex} and DP_{race} are the marginal diagnostics. Lower DP is better; **bold** marks the best mean per column.

Model	Accuracy $\uparrow$	DP $\downarrow$	DP_{sex} $\downarrow$	DP_{race} $\downarrow$
FADO	0.7979	0.0688	0.0234	0.0243
Aranyani (no controller)	0.8005	0.0727	0.0283	0.0294
ARF	0.7446	0.1497	0.0735	0.0620
RFR	0.7779	0.1427	0.0724	0.0591

Key findings

- **FADO more than halves the intersectional DP of both external baselines** ($\sim 54\%$ vs ARF, $\sim 52\%$ vs RFR). All twelve FADO-vs- $\{\text{ARF}, \text{RFR}\}$ comparisons are significant at $p < 0.001$ under *both* the paired Wilcoxon and the Welch test.
- **The controller now beats the controller-free baseline.** FADO reduces intersectional DP by 0.38pp over the bare Aranyani forest, significant under *both* tests (Wilcoxon $p < 0.001$, Welch $p < 0.01$), with matching wins on both marginal axes. In the single-attribute setting this comparison was only a statistical tie — so **the intersectional objective makes the controller’s contribution measurable**, at a negligible accuracy cost (-0.26pp , not significant under Welch).
- **Gerrymandering confirmed empirically.** The ratio of intersectional to worst marginal DP is $\approx 2\text{--}3$ for every method, and is *highest* for the fair learners (FADO ≈ 2.8 , Aranyani ≈ 2.5): they scrub the marginal axes so well that their residual unfairness is concentrated in the joint subgroups. A marginal-only audit would have declared FADO “very fair” while missing a joint-subgroup gap almost $3\times$ larger — direct empirical support for optimising the intersectional objective.
- **Statistics independently verified.** All significance values were cross-checked with two independent libraries (`scipy.stats` and `statsmodels.stats.multitest`); results match exactly.

4. Code quality: controller audit

I audited the full drift-reaction controller logic. Findings:

- Removed a dead, expensive ($O(T^2)$) fairness computation from the inner loop (measurable speed-up, no change to results).
- Identified three discrepancies between the documented design and the executed code (recovery driven by cumulative rather than rolling accuracy; the warning/confirmation ADWIN thresholds are inverted; the label-noise guard is weaker than described).

- **Decision:** these were left *unchanged* for now. On analysis, the current behaviour is plausibly what drives FADO’s advantage on the abrupt-drift regime, so “fixing” them risks erasing the contribution; they are documented for a controlled follow-up rather than a blind change.

5. Next steps (once the heatwave lets up)

- Finish the COMPAS intersectional 30-seed sweep.
- **Optimise equalised odds (EO) directly** (currently DP is the objective and EO is only monitored); the EO regulariser is implemented and driven by the same controller, so this is a configuration + sweep, then a second results block.
- Re-verify the gerrymandering and controller-gain claims hold on COMPAS.

A note on where I’d like to focus

One thing I’d genuinely like to invest more time in is a proper **ablation study of the reaction controller**. Right now it is the single weakest and most easily-targeted part of the work: a reviewer can reasonably ask “how much does each piece of the controller actually contribute, and would a simpler variant do just as well?” — and at the moment I don’t have a clean, systematic answer. Isolating each component (the dual-ADWIN trigger, the warning / confirmation / recovery phases, the rolling-window logic) and measuring its individual effect would let me answer that head-on and turn a likely point of criticism into a strength. I’m convinced this kind of ablation would meaningfully improve our odds at future conferences, even though it is time-consuming and won’t produce a headline result on its own. I’d appreciate the room to spend a bit of extra time here rather than rushing straight to the next dataset.